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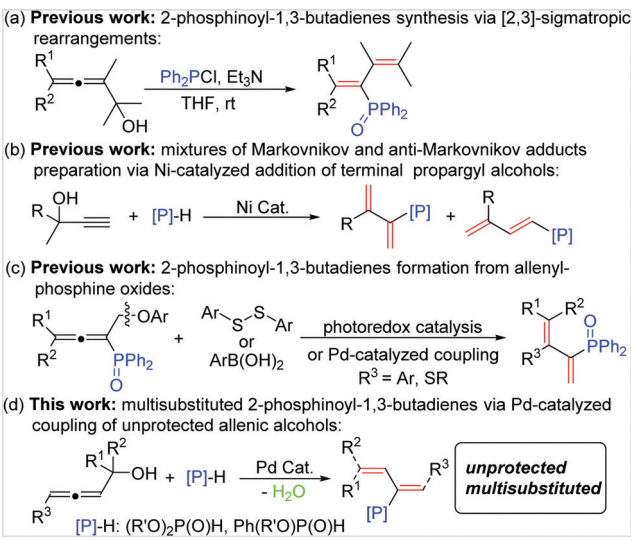
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Palladium-catalyzed C–P cross-coupling of allenic alcohols with *H*-phosphonates leading to 2-phosphinoyl-1,3-butadienes†

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The first facile, efficient, atom-economical and regioselective palladium-catalyzed direct C–P cross-coupling of unprotected allenic alcohols with *H*-phosphonates for the one-pot synthesis of structurally diverse multisubstituted 2-phosphinoyl-1,3-butadienes was developed. This strategy would enrich the allene chemistry and afford new scaffolds to construct complex molecular skeletons.

The construction of various kinds of organophosphorus compounds is a fundamental and significant topic in modern synthetic chemistry due to their broad applicability in biochemistry, pharmaceuticals, materials science and organic synthesis.[1] Phosphinoyl 1,3-butadienes as representative functionalized phosphorus compounds are an important class of butadiene-containing valuable precursors in synthetic chemistry, which could be readily transformed to other useful compounds such as allylic phosphonates obtained *via* Michael addition with nucleophiles,[2] alkenylphosphonates afforded through cycloaddition,[3] and vinyl allenes yielded through addition reaction with aldehydes.[4] Moreover, they are also potentially useful intermediates for P-ligand preparation.[5] Not surprisingly, considerable endeavor has been focused on the synthesis of 2-phosphinoyl-1,3-butadienes; however, the method for the preparation of these unique motifs, especially multisubstituted 2-phosphinoyl-1,3-butadienes, is quite rare to date and generally more challenging. In 1996, the procedure for the preparation of (1,3-dien-2-yl) diphenylphosphine oxides from allenic alcohols and air- and moisture-sensitive chlorodiphenylphosphine as raw materials was developed *via* [2,3]-sigmatropic rearrangements (Scheme 1a).[6] However, it heavily suffered from the use of unstable and toxic phosphorus chlorides, a large amount of bases, poor functional group compatibility and low yields, thus limiting its application. In 2005, Han's group disclosed the first nickel-catalyzed addition of P(O)H compounds to propargyl alcohols for the synthesis of 2-phosphinoyl-1,3-butadienes (Scheme 1b), but its general use poses severe limitations due to limited substrate scope and poor regioselectivity giving mixed products of Markovnikov and anti-Markovnikov adducts.[7] Recently, Pd-catalyzed Suzuki–Miyaura coupling of allenylphosphine oxides with arylboronic acids and the photoredox catalysis of allenylphosphine oxides with diaryl disulfides leading to (1,3-dien-2-yl) diphenylphosphine oxides were revealed (Scheme 1c).[8] However, these methods required the pre-preparation of complex and inaccessible allenylphosphine oxides as substrates and did not afford 1-substituted 2-phosphinoyl-1,3-butadienes. Therefore, new and economic methodologies for efficient access to highly



Scheme 1 Synthetic strategies for 2-phosphinoyl-1,3-butadienes.

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† Electronic supplementary information (ESI) available: General experimental procedures, characterization details and copies of ^{1}H, ^{13}C, and ^{31}P NMR spectra of compounds. CCDC 2034498. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0cc07022j

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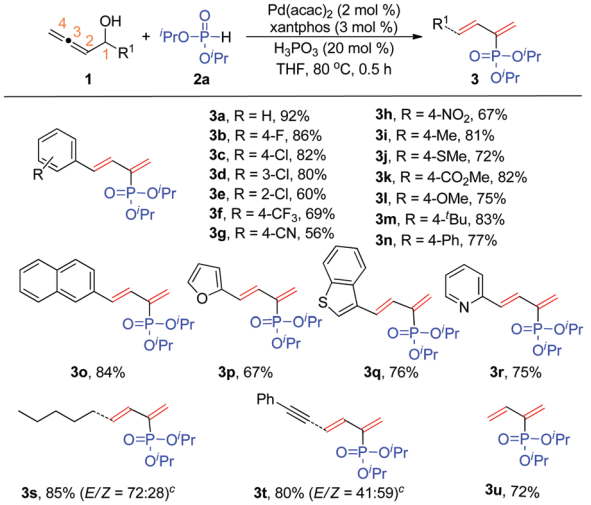
substituted 2-phosphinoyl-1,3-butadienes with broad substrate compatibility and excellent regioselectivity are still desired.

Allenic alcohols, an important class of allene compounds with a distinctive substructure of a hydroxyl group and two cumulative carbon–carbon double bonds, are readily available[9] and versatile building blocks for making valuable molecular skeletons, such as dihydrofurans,[10] vinylic oxiranes,[11] unsaturated ketones,[12] and 1,3-butadienes.[13] In addition, from economic and environmental points of view, the direct use of unprotected allenic alcohols as starting materials is highly attractive for modern organic synthesis as water is the only byproduct in some transformations[13b,14] and due to their prominent advantages of step and atom economy. Nevertheless, to the best of our knowledge, no example of 2-phosphinoyl-1,3-butadiene synthesis *via* transition-metal-catalyzed direct coupling of allenic alcohols with P(O)H compounds was reported. As part of our ongoing endeavor to develop environmentally benign novel approaches for P–C bond formation,[15] herein, we demonstrate the first example of Pd-catalyzed direct cross-coupling of various allenic alcohols with *H*-phosphonates to access structurally diverse 2-phosphinoyl-1,3-butadienes, providing a quite convenient and efficient tool to construct a wide range of multisubstituted 2-phosphinoyl butadiene frameworks along with the production of new P–C, C–C and C=C bonds using a simple one-pot process (Scheme 1d).

Initially, the model reaction of 1-phenylbuta-2,3-dien-1-ol (**1a**) and diisopropyl phosphonate (**2a**) was conducted to find the optimal reaction conditions, including catalysts, ligands, additives, solvents and temperatures. The results revealed that the optimal conditions were as follows: 2 mol% $Pd(acac)_2$ as the catalyst, 3 mol% xantphos as the ligand, 20 mol% H_3PO_3 as the additive and tetrahydrofuran (THF) as the solvent at 80 °C under argon for 0.5 h with 1.1: 1 ratio of **1a**/**2a** (see Table S1 in the ESI† for details).

With the optimized reaction conditions in hand (Table S1, entry 19, ESI†), the substrate scope of Pd-catalyzed coupling of structurally diverse 1-substituted allenic alcohols with **2a** was first investigated. As shown in Table 1, the method was found to be quite general and various 2-phosphinoyl-1,3-butadienes could be conveniently and efficiently synthesized using this facile and novel coupling. Generally, a wide range of 1-aryl buta-2,3-dien-1-ols decorated with electron-donating groups and electron-withdrawing groups on the phenyl ring, such as halides (**1b–1e**), trifluoromethyl (**1f**), cyano (**1g**), methyl (**1i**), methylthio (**1j**), ester (**1k**), methoxyl (**1l**), *tert*-butyl (**1m**) and phenyl (**1n**), could be transformed into the corresponding coupling products (**3a–3n**) in moderate to excellent yields, without a distinct electronic effect on the reactivity being observed. It is noteworthy that high selectivity for *E* stereoisomers of the above-mentioned products (**3a–3r**) was observed owing to the steric hindrance on the basis of *in situ* ^{31}P and ^{1}H NMR spectroscopic analyses,[7,16] and the relative trace *Z* stereoisomers were not isolated using column chromatography on silica gel. To be noted, the catalytic efficiency was not altered by changing the substituent R^1 to *n*-pentyl and phenylethynyl (**1s** and **1t**), but the corresponding products **3s** and **3t** were obtained as a mixture of *E*/*Z* isomers in high total yields.

Table 1 Palladium-catalyzed coupling reaction of 1-substituted allenic alcohols with **2a**[abc]



[a] Reaction conditions: **1** (0.33 mmol), **2a** (0.30 mmol), $Pd(acac)_2$ (2 mol%), xantphos (3 mol%) and H_3PO_3 (20 mol%) in THF (4.5 mL) at 80 °C for 0.5 h under argon. [b] Isolated yield. [c] *E*/*Z* ratio was obtained using ^{31}P NMR and ^{1}H NMR spectroscopic analyses.

Interestingly, the simple buta-2,3-dien-1-ol **1u** was also suitable for the coupling, providing **3u** in 72% yield.

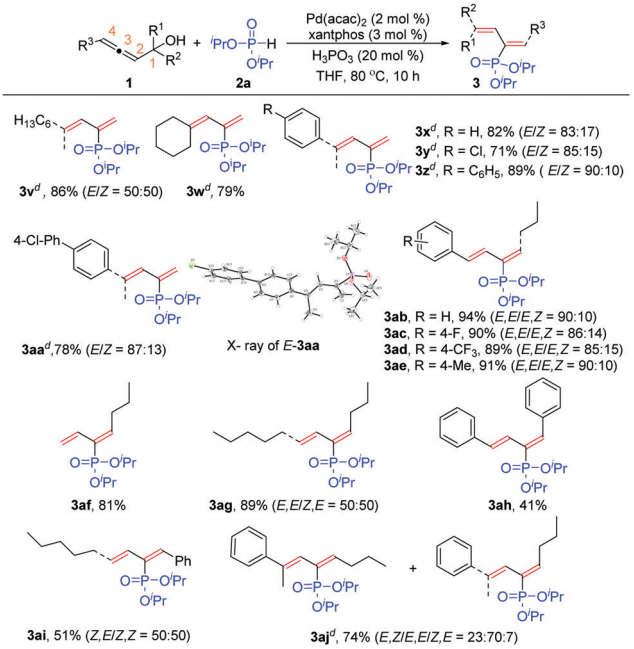
To further verify the generality of this new coupling to produce 2-phosphinoyl-1,3-butadienes, multisubstituted allenic alcohols including 1,2-disubstituted, 1,4-disubstituted and 1,1,4-trisubstituted allenic alcohols were explored, and the results are summarized in Table 2. Various tertiary 2,3-allenols (**1v–1aa**) could be successfully engaged in this reaction, affording the corresponding products **3v–3aa** in high total yields by enhancing the temperature and the loading of the additive along with extending the reaction time. Fortunately, the structural assignment was corroborated using X-ray crystallography of product **3aa** (CCDC† 2034498),[17] which clearly disclosed that the phosphinoyl moiety was preferentially installed at the C3-position of **1aa** in this coupling. We next investigated the effect of substitutions at the terminal position of the allene moiety on the yield (**1ab–1aj**). When 1-aryl-4-alkyl 2,3-allenols (**1ab–1ae**) bearing F, CF_3 and CH_3 groups were employed, excellent total yields of 89–94% with a high *E*/*Z* ratio were obtained. Additionally, alkyl-substituted allenic alcohols (**1af** and **1ag**) could also participate in the coupling to afford the target products **3af** and **3ag**, respectively, in high yields. Yet, the allenic alcohols bearing a phenyl group at the C4 position (**1ah** and **1ai**) led to a marked decrease of the coupling yield (**3ah**, 41%; **3ai**, 51%). To our delight, a 1,1,4-trisubstituted allenic alcohol **1aj** was proven to be viable for this coupling. Furthermore, it was noteworthy that the allenic alcohols with an aryl moiety at the C1 position exhibited higher stereoselectivity for *E*-isomers over those containing an alkyl group owing to the effect of steric hindrance.

To further extend the scope of this reaction, some other P(O)H substrates were then studied (Table 3). In addition to **2a**, other *H*-phosphonates such as dimethyl (**2b**), diethyl (**2c**),

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Table 2 Pd-catalyzed coupling reaction of multisubstituted allenic alcohols with **2a**[abc]



[a] Reaction conditions: **1** (0.33 mmol), **2a** (0.30 mmol), $Pd(acac)_2$ (2 mol%), xantphos (3 mol%) and H_3PO_3 (20 mol%) in THF (4.5 mL) at 80 °C for 10 h under argon. [b] Isolated yield. [c] *E*/*Z* ratio was obtained using ^{31}P and ^{1}H NMR spectroscopic analyses. [d] H_3PO_3 (60 mol%), 90 °C for 10 h.

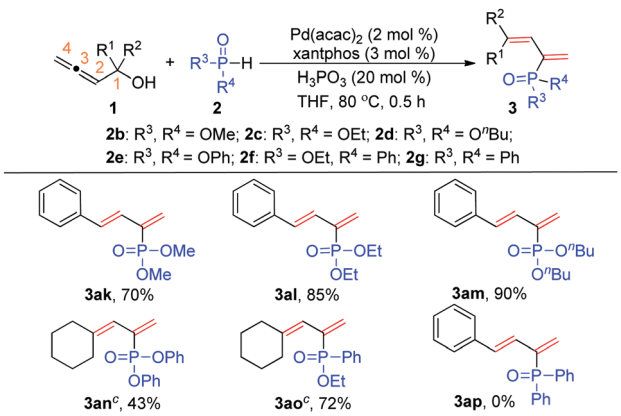
dibutyl (**2d**) and diphenyl (**2e**) phosphonates could be used as the suitable partners, generating the corresponding products (**3ak–3an**) in moderate to excellent yields. To our satisfaction, an *H*-phosphinate ester **2f** proceeded smoothly to provide a 72% yield of product **3ao**. However, the secondary phosphine oxide **2g** was inert under similar reaction conditions, which might be attributed to the electronic properties of P(O)H compounds.

To demonstrate the synthetic application of the present method, a scaled-up reaction and further transformations of some products were performed (Scheme 2). A gram-scale synthesis of **3u** was conducted by using 20 mmol of **2a** under similar reaction conditions, producing **3u** in 65% yield (Scheme 2a). Moreover, 2-phosphinoyl-1,3-butadiene products could be easily transformed to other useful compounds *via* Diels–Alder reaction (Scheme 2b). As expected, 2-phosphinoyl-1,3-butadienes like **3u** could undergo Michael addition with common nucleophiles such as **2i** and **2g** to generate the corresponding products **5a** and **5b** in 77% and 51% yields, respectively (Scheme 2c).

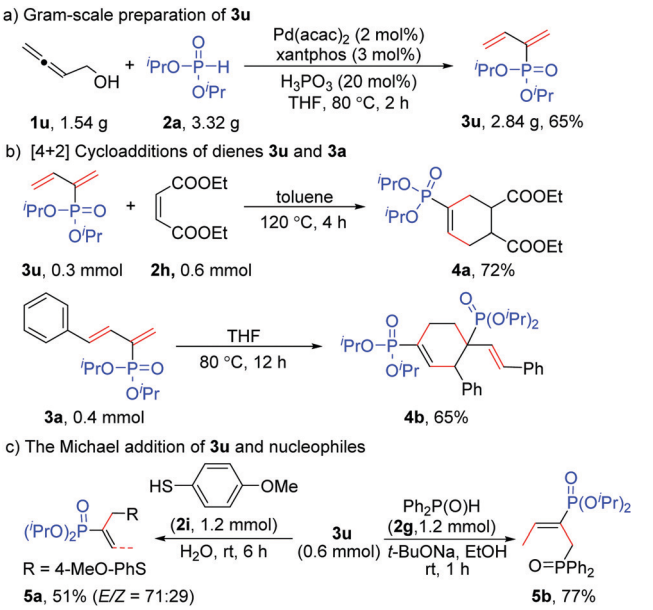
Based on the experimental facts and related literature reports,[18] a plausible mechanism is presented in Scheme 3. Initially, substrate **1a** was activated through a hydrogen bond interaction with phosphorous acid to generate the reactive species **A**.[18a] Subsequently, palladium(0) attacked the C3-position of **A** along with the cleavage of the C–O bond, leading to the formation of allylpalladium hydroxide **B**, which was then subjected to transmetalation with **2a** (in the form of the trivalent phosphine oxide **2a′**) to result in intermediate **C**.[18b–d] Finally, the reductive elimination of palladium species **C** toward C–P bond formation occurred to give the coupling product **3a** and regenerated the palladium(0) catalyst.[18b,e]

In conclusion, we have documented the first facile and efficient palladium-catalyzed direct cross-coupling reaction of easily accessible allenic alcohols with various *H*-phosphonates, furnishing a new and convenient route to synthesize a structurally diverse array of novel multisubstituted 2-phosphinoyl-1,3-butadienes along with the concurrent formation of new C–C, C=C, and C–P bonds using a simple one-step process. The obtained coupling products can be easily transformed into

Table 3 Substrate scope of P(O)H compounds[ab]

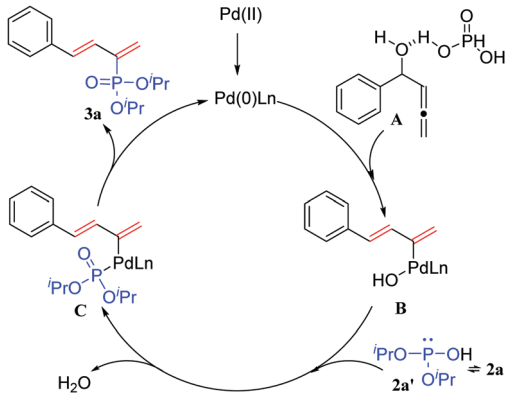


[a] Reaction conditions: **1** (0.33 mmol), **2** (0.30 mmol), $Pd(acac)_2$ (2 mol%), xantphos (3 mol%) and H_3PO_3 (20 mol%) in THF (4.5 mL) at 80 °C for 0.5 h under argon. [b] Isolated yield. [c] H_3PO_3 (60 mol%), 90 °C for 10 h.



Scheme 2 Application studies.

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Scheme 3 Plausible reaction mechanism.

useful multifunctionalized phosphonates *via* some simple reactions. The salient features of this protocol are the excellent chemo- and site selectivity, operational simplicity, remarkable functional group tolerance, high atom- and step-economy and good to excellent yields. Thus, we believe that this newly developed approach would enrich the allene chemistry and provide new scaffolds to construct valuable and complex molecular skeletons.

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Conflicts of interest

There are no conflicts to declare.

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